MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, COAXIAL, 0.141 DIAMETER, SEMIRIGID, 50 OHMS, M17/130-RG402 AND M17/130-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cables described herein shall consist of this document and the latest issue of Specification MIL-C-17.

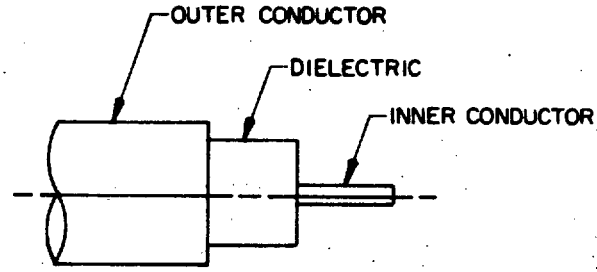


FIGURE 1. Configuration.

TABLE I. Description.

Part number	Inner conductor	Dielectric core	Outer conductor
M17/130-RG402	Solid, silver-coated copper-clad steel Diameter, .0362 ±0007 inch	Type F-1 Diameter, .1175 inch ±.0010	Seamless copper tubing Diameter, .141 inch ±.001
M17/130-00001	Same as above	Same as above	Seamless copper tubing, Diameter, .141 inch +.002, -.001 Tin plated per MIL-T-10727, Type I, .0003 to .0005-inch thick.

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ENGINEERING INFORMATION:

Continuous working voltage: 1,900 Vrms, maximum.
Operating frequency: 20 GHz, maximum.
Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -40° to +100°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 2.05 ohms per 100 feet.
Elongation: 10 percent minimum for class 40 A wire.
1.0 percent minimum for class 40 HS wire.
Tensile strength: 45 klb/inch^2 minimum for class 40 A wire.
110 klb/inch^2 minimum for class 40 HS wire.
Engineering notes: This cable useful in critical RF performance applications (see connector series SMA per MIL-C-39012). This cable manufactured in 20-foot lengths.

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Eccentricity: 7.5 percent, maximum.
Adhesion of conductors:
Inner conductor to core: 4 pounds minimum; 55 pounds maximum.
Aging stability: Not applicable.
Stress crack resistance: Not applicable.
Outer conductor integrity: +175° ±5°C.
Cold bend: Not applicable.
Outer conductor integrity: +175° ±5°C.
Dimensional stability: +125° ±5°C.
Core to jacket: .015 inch, maximum.
Contamination: Not applicable.
Bendability: Mandrel diameter .50 inch, maximum.
Flammability: Not applicable.
Weight: 34.4 pounds per 100 feet maximum (M17/130-RG402).
35.1 pounds per 100 feet maximum (M17/130-00001).

Electrical:
Test frequency: 500 MHz to 20 GHz.
Spark test: Not applicable.
Voltage withstanding: 5,000 Vrms.
Insulation resistance: Not applicable.
Corona extinction voltage: 1,900 Vrms, minimum.
Characteristic impedance: 50.0 ohms ±1.
Attenuation: See figure 2.
Structural return loss: See figure 3.
Capacitance: 29.9 pF per foot, maximum.
Capacitance stability: Not applicable.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Phase stability: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

Part number: See table II.

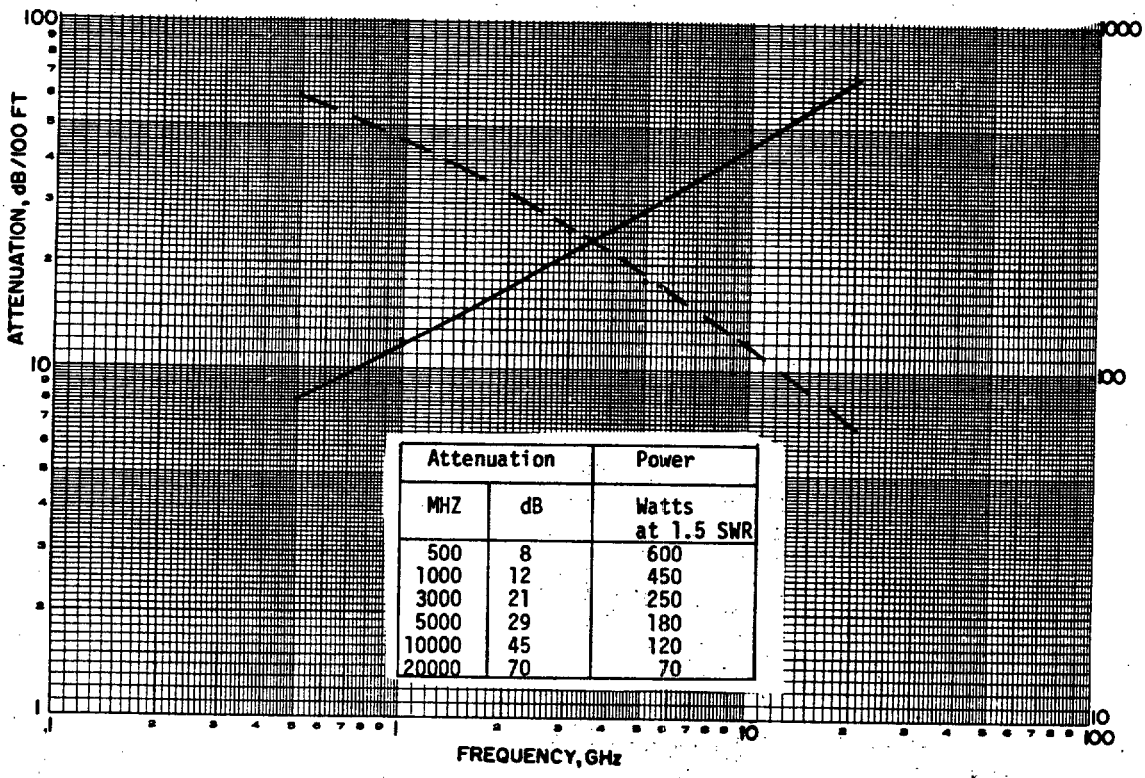
Supersession data: See table II.

TABLE II. Cross reference of part number.

Part number	Superseded part number or type designation
M17/130-RG402	RG-402/U
M17/130-00001	---

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

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Attenuation		Power
MHZ	dB	Watts at 1.5 SWR
500	8	600
1000	12	450
3000	21	250
5000	29	180
10000	45	120
20000	70	70

MAXIMUM ATTENUATION —— (Test requirements shall be as noted as line indicated on graph).
MAXIMUM POWER – – –
AT 25°C SEA LEVEL

FIGURE 2. Power rating and attenuation.

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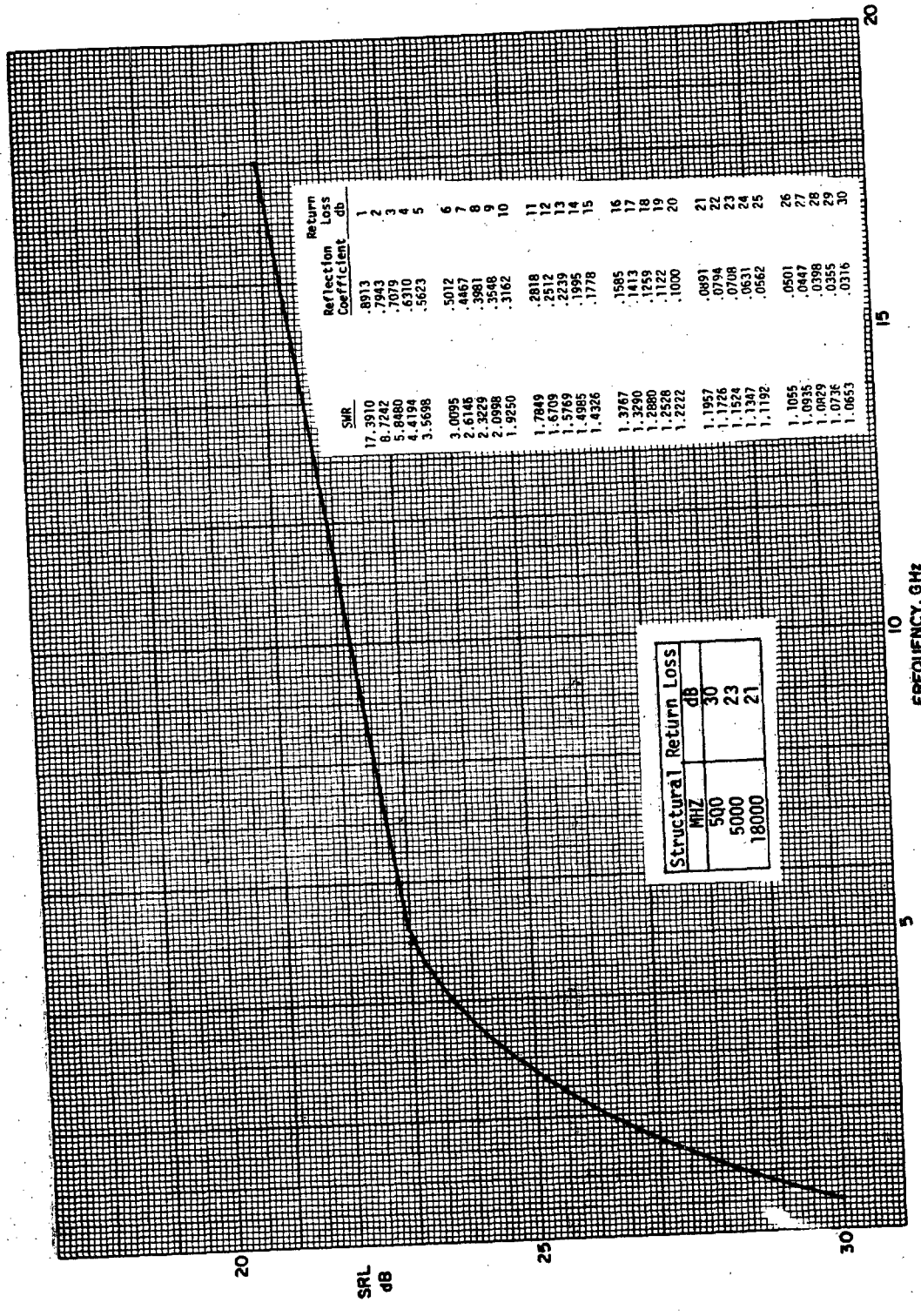


FIGURE 3. Minimum structural return loss cable.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - ME, AT, AR
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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